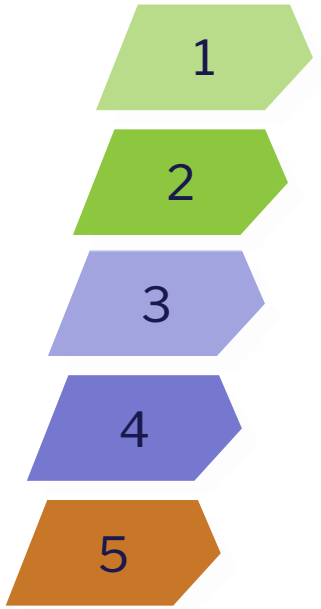
Rack and Power Update

Harry Soin Senior Director Advanced Energy

Steven Moore Director Rittal

Outline



Rack and Power Update

Hyperscale Power architecture migration

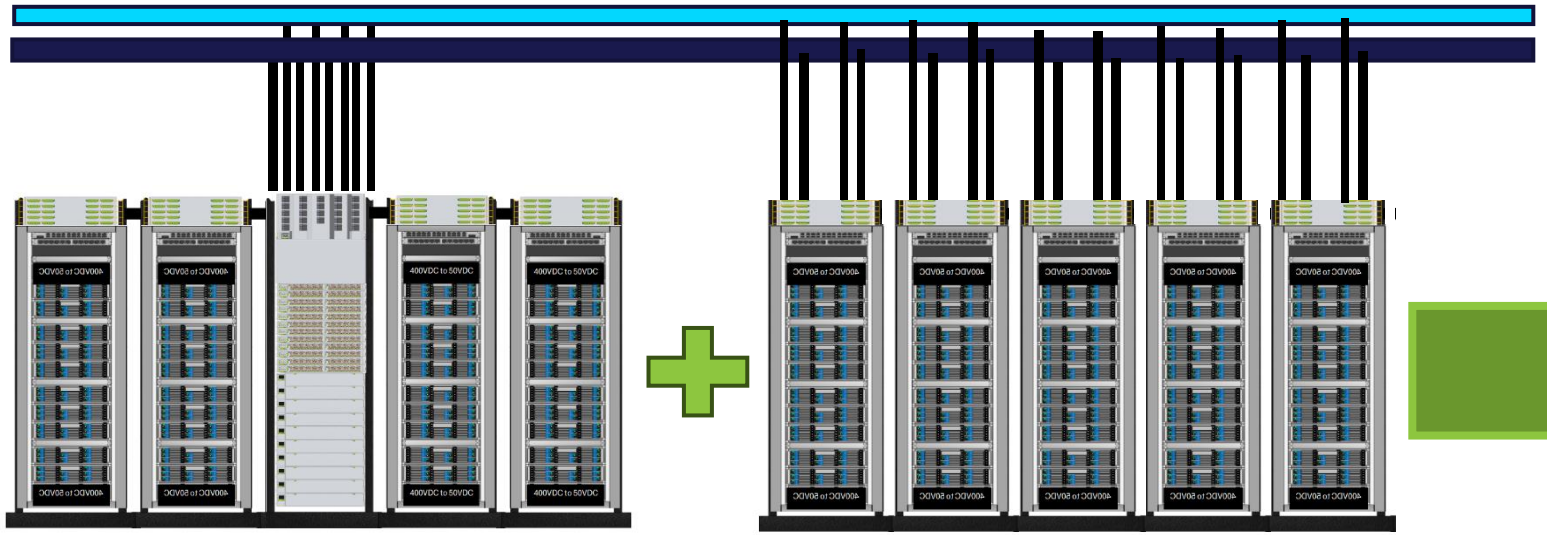
Status Update

What is new

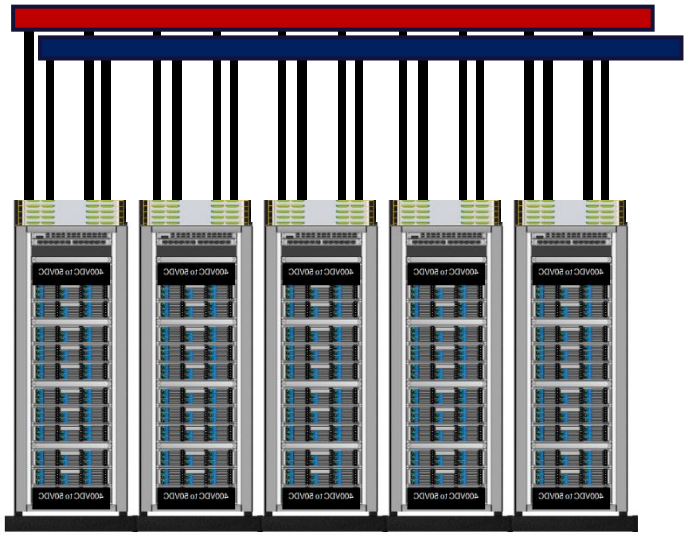
Call for Action

Hyperscale Power Architecture Migration

AC Busway(s) 415/480VAC



LVDC Busway(s)



LVDC Sidecar

Power + Optional BBU

50V bus in IT racks

Mt. Diablo

High power density shelves

3-phase PSUs in Power shelves

liquid cooled Power shelves

Infra level LVDC

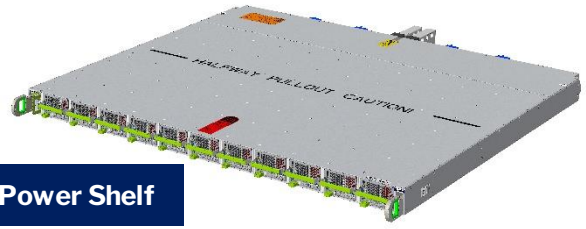
LVDC Distribution

Direct to server Power

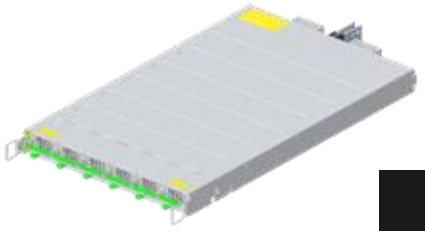
End of row SST

2025 **2027** **2028+**

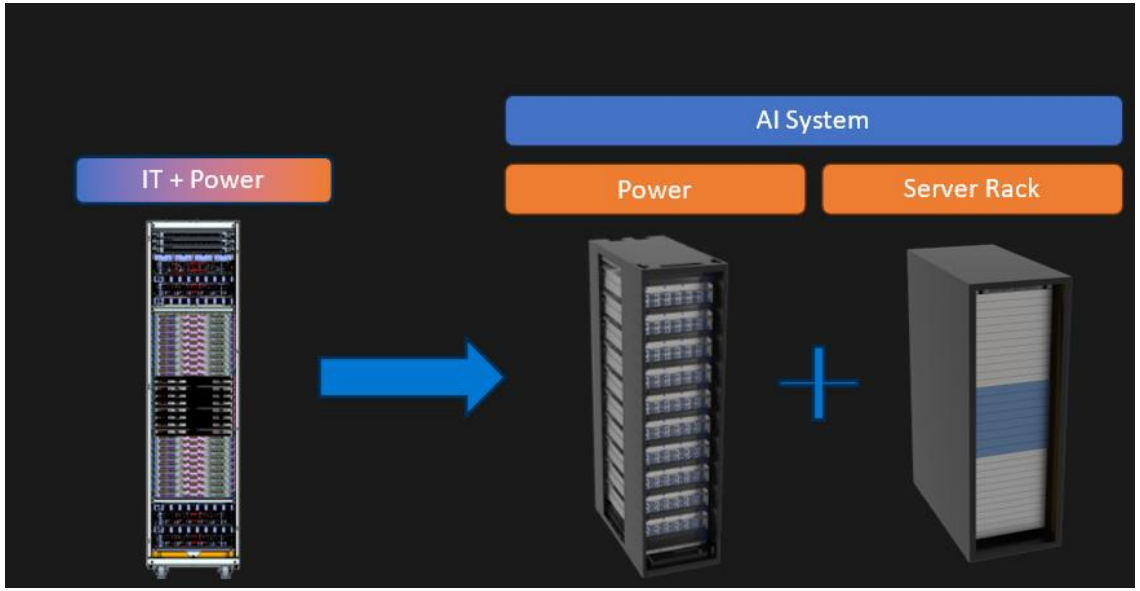
OCP Rack Power Growth Drivers



HPRv4 160kW Power Shelf



HPRv2 72kW Power Shelf





HPRv1 33kW Power Shelf



ORv3 18kW Power Shelf



HPRv4 100kW Power Shelf

Status Update

Highlights from this year

- OR-W Specifications and design file updates
- MGX Accelerated Computing Rack Specifications
- Power Projections by Nvidia
- Mount Diablo Specifications
- TE Liquid cooling busbar updates
- HRPV2 Power Supply and Power shelf updates
- 100kW/160kW HPRv4 Power shelf and 18kW DC-DC PSU updates
- DMTF updates : Redfish profiles

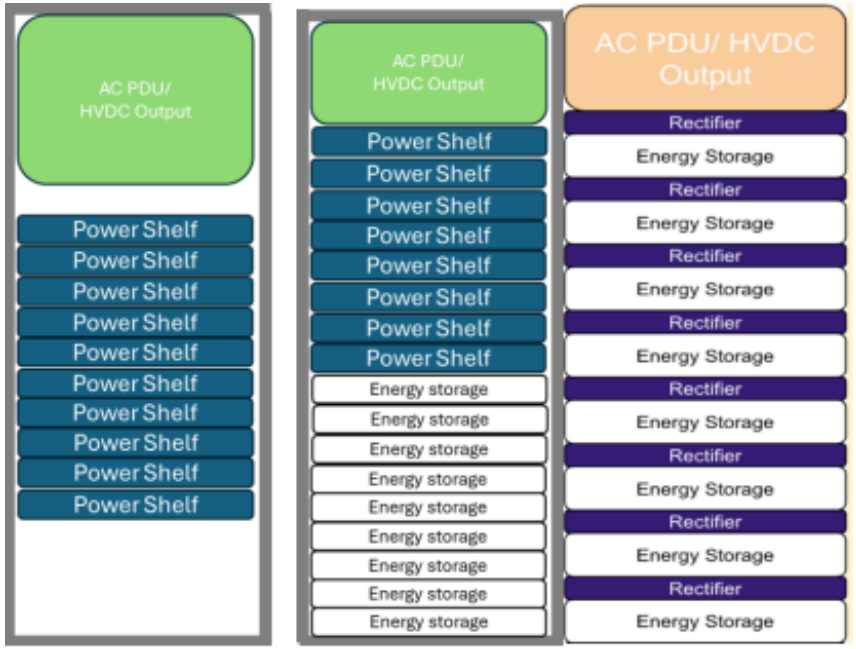


Figure 1: Three examples of power rack configurations.

Roadmap

Plans for upcoming year

- 3 Phase architectures
- LVDC architectures - 800V & +/-400V.
  - High Power connectors and cabling options
  - LVDC to server architectures
  - Safety aspects
  - Racks and busbar specifications

- Mount Diablo update
- OR-W Rack Development
- Joint Discussion with Immersion work stream
- Redfish Power schema for High Power

New Sub Project



LVDC Power Delivery Sub Project

Scope: standardizing LVDC power delivery from facility input through the power rack to the IT load, covering both converted and native delivery

Workstreams: Converted and Native LVDC Delivery, co-leads from Meta, Google, Microsoft and Nvidia
Sub-project leads: Zuhir Sras (Meta), David Sun (Google)

Call to action: connector, busbar and power shelf suppliers, integrators, and other hyperscalers. **First call November**

Call to Action

- Get involved in the Project/Sub-Project Community!
- Involvement within the community is key for the success of next generation power specifications.
- Defining individual requirements are key to creating an overall industry specification.

- Project Wiki with latest specification : https://www.opencompute.org/wiki/Open_Rack/SpecsAndDesigns
- Mailing list: https://www.opencompute.org/projects/rack-and-power